

Nitrogen

N (N I). Strongest classified transitions below 2 μm (NIST): 69 levels, 98 lines.

N⁺ ionisation limit 117225.70 cm⁻¹ = 14.53413 eV (85.306 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U|er|U\rangle$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock; M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



non-uniform scale

cm⁻¹

Energy above the ground state

cm⁻¹

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Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p ³ ⁴ S _{3/2} - 2s ² 2p ³ ² P _{3/2}	346.7490	346.6497	864.5806	-	6.500e-03	1.511e-08	6.84	auroral [N I] 346.6 nm line
2s ² 2p ³ ⁴ S _{3/2} - 2s ² 2p ³ ² P _{1/2}	346.7536	346.6543	864.5691	-	2.600e-03	1.436e-08	2.88	auroral [N I] 346.6 nm line
2s ² 2p ³ ⁴ S _{3/2} - 2s ² 2p ³ ² D _{3/2}	519.9349	519.7902	576.5961	-	1.600e-05	3.239e-12	78.6	auroral / nightglow / nebular [N I] 520 nm doublet
2s ² 2p ³ ⁴ S _{3/2} - 2s ² 2p ³ ² D _{5/2}	520.1705	520.0257	576.3349	-	6.590e-06	1.203e-12	87.2	auroral / nightglow / nebular [N I] 520 nm doublet
2s ² 2p ³ 3s ⁴ P _{1/2} - 2s ² 2p ³ 3p ⁴ S _{3/2}	742.5686	742.3641	403.7236	2.148	5.710e+06	6.121	14.8	742.57 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ S _{5/2}	744.4348	744.2298	402.7115	3.104	1.183e+07	6.121	30.8	744.43 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ S _{7/2}	747.0369	746.8312	401.3088	3.839	1.791e+07	6.121	46.6	747.04 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ P _{5/2}	818.7112	818.4861	366.1761	3.574	7.860e+06	2.894	43.2	818.71 nm
2s ² 2p ³ 3s ⁴ P _{1/2} - 2s ² 2p ³ 3p ⁴ P _{3/2}	819.0263	818.8012	366.0352	3.638	1.220e+07	-	-	819.03 nm
2s ² 2p ³ 3s ⁴ P _{1/2} - 2s ² 2p ³ 3p ⁴ P _{1/2}	820.2611	820.0357	365.4842	1.625	4.850e+06	-	-	820.26 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ P _{3/2}	821.2972	821.0715	365.0231	2.343	5.020e+06	-	-	821.30 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ P _{5/2}	821.8595	821.6336	364.7734	6.027	2.210e+07	2.894	100	821.86 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ P _{1/2}	822.5389	822.3128	364.4721	3.815	2.650e+07	-	-	822.54 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ P _{5/2}	824.4655	824.2389	363.6204	3.850	1.340e+07	-	-	824.47 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ D _{3/2}	868.2666	868.0282	345.2770	7.743	2.320e+07	3.617	100	868.27 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ D _{5/2}	868.5788	868.3403	345.1529	5.792	1.729e+07	-	-	868.58 nm
2s ² 2p ³ 3s ⁴ P _{1/2} - 2s ² 2p ³ 3p ⁴ D _{3/2}	868.8535	868.6149	345.0437	3.717	1.067e+07	-	-	868.85 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ D _{5/2}	870.5637	870.3247	344.3659	3.609	2.000e+07	-	-	870.56 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ D _{5/2}	871.4096	871.1703	344.0316	3.966	1.204e+07	-	-	871.41 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ D _{1/2}	872.1232	871.8837	343.7501	3.422	5.960e+06	-	-	872.12 nm
2s ² 2p ³ 3s ⁴ P _{5/2} - 2s ² 2p ³ 3p ⁴ D _{1/2}	873.1209	872.8901	343.3538	1.559	3.700e+06	-	-	873.13 nm
2s ² 2p ³ 3s ⁴ P _{3/2} - 2s ² 2p ³ 3p ⁴ D _{3/2}	874.9771	874.7369	342.6289	1.079	8.800e+05	-	-	874.98 nm
2s ² 2p ³ ² D _{3/2} - 2s ² 2p ³ ² P _{3/2}	1040.0587	1039.7737	288.2457	-	6.000e-02	1.511e-08	63.2	[N I] 1040 nm infrared auroral multiplet
2s ² 2p ³ ² D _{5/2} - 2s ² 2p ³ ² P _{1/2}	1040.1004	1039.8155	288.2341	-	3.450e-02	1.436e-08	38.2	[N I] 1040 nm infrared auroral multiplet

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴N hyperfine structure (I = 1)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2s ² 2p ³ ⁴ S _{3/2}	10.450929	0.000001	F=1/2: -26.13 F=3/2: -10.45 F=5/2: +15.68
2s ² 2p ³ 3s ⁴ P _{1/2}	112.3(13)	-	F=1/2: -112.30 F=3/2: +56.15
2s ² 2p ³ 3s ⁴ P _{3/2}	68.3(7)	4(9)	F=1/2: -166.45 F=3/2: -71.83 F=5/2: +103.37
2s ² 2p ³ 3s ⁴ P _{5/2}	129.5(8)	-7.8(20)	F=3/2: -458.78 F=5/2: -123.28 F=7/2: +321.85
2s ² 2p ³ 3p ⁴ D _{3/2}	69.8(9)	-	F=1/2: -69.76 F=3/2: +34.88
2s ² 2p ³ 3p ⁴ D _{5/2}	30.3(15)	-0.9(17)	F=1/2: -76.88 F=3/2: -29.40 F=5/2: +45.23
2s ² 2p ³ 3p ⁴ D _{1/2}	36.6(11)	-4.1(25)	F=3/2: -130.97 F=5/2: -33.32 F=7/2: +90.47
2s ² 2p ³ 3p ⁴ D _{3/2}	55.2(11)	-9.9(26)	F=5/2: -253.70 F=7/2: -48.13 F=9/2: +190.72
2s ² 2p ³ 3p ⁴ P _{1/2}	-50.8(17)	-	F=1/2: +50.78 F=3/2: -25.39
2s ² 2p ³ 3p ⁴ P _{3/2}	46.2(15)	-2.7(17)	F=1/2: -118.88 F=3/2: -43.50 F=5/2: +68.62
2s ² 2p ³ 3p ⁴ P _{5/2}	31.9(9)	1.1(21)	F=3/2: -110.98 F=5/2: -32.81 F=7/2: +80.10

Measured A, B constants; sources in the data file.